

3D Reconstruction from Spherical Images: A Review of Techniques, Applications, and Prospects

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Abstract—3D reconstruction plays an increasingly important role in modern photogrammetric systems. Conventional satellite or aerial-based remote sensing (RS) platforms can provide the necessary data sources for the 3D reconstruction of large-scale landforms and cities. Even with low-altitude UAVs (Unmanned Aerial Vehicles), 3D reconstruction in complicated situations, such as urban canyons and indoor scenes, is challenging due to frequent tracking failures between camera frames and high data collection costs. Recently, spherical images have been extensively used due to the capability of recording surrounding environments from one camera exposure. In contrast to perspective images with limited FOV (Field of View), spherical images can cover the whole scene with full horizontal and vertical FOV and facilitate camera tracking and data acquisition in these complex scenes. With the rapid evolution and extensive use of professional and consumer-grade spherical cameras, spherical images show great potential for the 3D modeling of urban and indoor scenes. Classical 3D reconstruction pipelines, however, cannot be directly used for spherical images. Besides, there exist few software packages that are designed for the 3D reconstruction from spherical images. As a result, this research provides a thorough survey of the state-of-the-art for 3D reconstruction from spherical images in terms of data acquisition, feature detection and matching, image orientation, and dense matching as well as presenting promising applications and discussing potential prospects. We anticipate that this study offers insightful clues to direct future research.

Index Terms—spherical image; equirectangular projection; 3D reconstruction; structure from motion; simultaneous localization and mapping; dense matching; image matching

I. INTRODUCTION

3D reconstruction is an increasingly critical module in recent photogrammetric systems. It has been extensively utilized for constructing digital cities [1], documenting cultural heritages [2], and inspecting tunnel cracks [3], etc. 3D reconstruction can be implemented by using varying instruments, e.g., LiDAR (Light Detection and Ranging) scanners, TOF (Time of Flight) sensors, and optical cameras. The popularity of image sensors and the development of processing techniques have led to the vast usage of image-based 3D reconstruction techniques among all available sensors in the field of photogrammetry and remote sensing (RS), such as satellite and aerial-based images for urban buildings [4].

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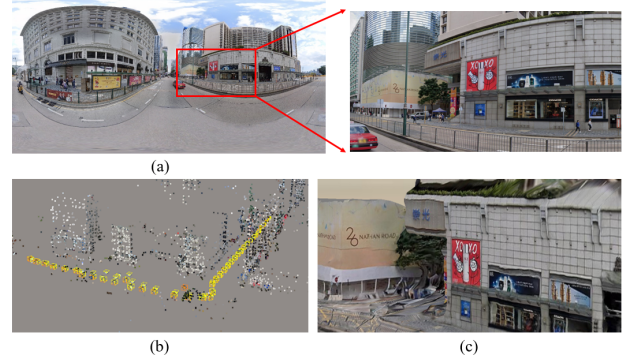


Fig. 1. 3D reconstruction based on spherical images. (a) a spherical image and one enlarged region for detail comparison; (b) the sparse reconstruction model; and (c) the dense reconstruction model after texture mapping.

Due to the relatively high acquisition altitudes and nearly vertical viewpoints, satellite and aerial-based RS images mainly observe building roofs over large-scale regions. With the increasing demands for fine-scale modeling, such as building facades and indoor environments, recent years have witnessed explosive development of 3D reconstruction based on low-altitude unmanned aerial vehicles (UAV) [71], [140] or terrestrial mobile mapping systems (MMS) [158]. Compared with satellite and airborne-based RS platforms, these near-ground platforms have the advantages of flexible instrument integration and multi-view imaging, which can record the contents that cannot be observed from high altitudes. Therefore, UAV and MMS have been used as essential RS platforms for data acquisitions in urban and indoor scenes [24].

Perspective cameras are the most widely used sensors for image-based 3D reconstruction. However, due to the characteristics of data acquisition in street-view and indoor environments, two main issues occur for perspective cameras with their limited FOV (Field of View) [5]. On the one hand, surrounding environments are different from aerial photogrammetry. In street-view and indoor environments, the trajectory of data acquisitions is limited by street structures and indoor layouts, which causes sudden viewpoint changes at turning points and track failure between camera frames [6]. On the other hand, the observation regions are extended from the single-direction records in aerial photogrammetry to the omnidirectional acquisitions in street-view and indoor environments. It requires more images at each camera exposure position and increases acquisition time consumption [7]. Thus, effective imaging techniques are needed for 3D reconstruction in street-view and indoor situations.

Spherical cameras, also termed 360 cameras or omnidirectional cameras, can record all surrounding environments using one camera exposure. In contrast to traditional perspective cameras, recorded images of spherical cameras can cover the whole scene, whose FOV ranges are 360 and 180 degrees in horizontal and vertical directions, respectively. Due to the advantage of spherical cameras, spherical images have been adopted for 3D modeling in street-view and indoor environments [8], [9]. In addition, low-cost consumer-grade spherical cameras like the Insta360 and Ricoh theta [10] are growing in popularity, which greatly simplifies data acquisition and encourages their use in a variety of fields, such as damaged building inspection [11], urban environment analysis [12], urban geo-localization [13], [14], and heritage modeling [15]. Thus, spherical images have become one of the important data sources for 3D reconstruction, especially for street-view and indoor environments [16], as illustrated in Fig. 1.

Spherical images, however, have different characteristics when compared with traditional perspective images in the context of image-based 3D reconstruction [17]. One of the most important differences is the camera imaging model. Consequently, 3D reconstruction from spherical images has technique differences from perspective images. In addition, fewer commercial and open-source solutions are designed for the 3D reconstruction from spherical images when compared with perspective images. Therefore, this study aims to give a review of reported techniques related to 3D reconstruction from spherical images. The main contributions of this study include: (1) we give a systematic and extensive review of recent techniques for 3D reconstruction from spherical images; (2) we present the most promising applications related to 3D reconstruction from spherical images; and (3) we also conclude the prospects for 3D reconstruction from spherical images from the aspects of technique development and application promotion. The purpose of this study is to provide useful clues to guide further research for 3D reconstruction from spherical images.

This paper is organized as follows. The state-of-the-art of data acquisition, image matching, image orientation, and dense matching for 3D modeling of spherical images is reviewed in Section II. Section III examines prospective applications for 3D reconstruction from spherical images, which is followed by the primary prospects presented in Section IV. Finally, Section V concludes this work and future studies.

II. TECHNIQUES

This section presents the recent techniques for 3D reconstruction from spherical images. The main workflow for the photogrammetric 3D reconstruction is first introduced, followed by data acquisition with varying spherical cameras, image matching for establishing correspondences, image orientation for estimating camera poses, and dense matching for producing point clouds. The details are listed as follows.

A. Main workflow

According to the literature, there are five significant steps in the workflow of image-based 3D reconstruction, i.e., data

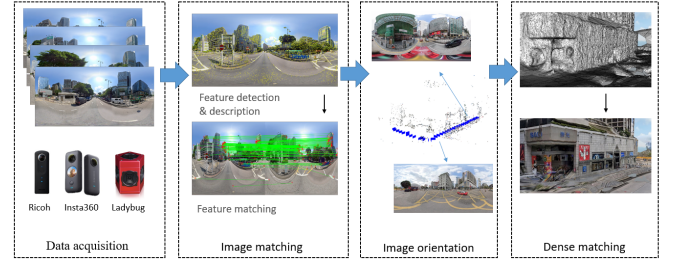


Fig. 2. The main workflow of 3D reconstruction for spherical images.

acquisition using photogrammetric systems, image matching to establish correspondences, image orientation to determine camera poses, dense matching to produce point clouds, and point cloud meshing and texturing. Spherical images differ in the camera imaging model and image representation format compared with classical perspective images, which causes extra considerations in the first four steps. Thus, this study reviews data acquisition, image matching, image orientation, and dense matching, and the main workflow of 3D reconstruction for spherical images is presented in Fig. 2.

B. Data acquisition

Data acquisition is the first step in the main workflow of 3D reconstruction for spherical images. In this section, three topics related to data acquisition are presented, i.e., spherical cameras, image representations, and acquisition platforms. The details are shown in the following subsections.

1) *Spherical camera*: The development of spherical cameras can be traced back to two centuries ago [18], which are invented for the documentation of ancient buildings and cultural heritages. For the purpose of surveying and mapping, spherical cameras were first used in close-range photogrammetry, in which spherical images were usually captured by rotating camera around the projection center or stitching overlapped images through image matching. For aerial photogrammetry, spherical cameras are designed as an integrated instrument that consists of several well-calibrated perspective cameras.

In recent years, the performance of spherical cameras has been greatly improved by the progress in the fields of camera

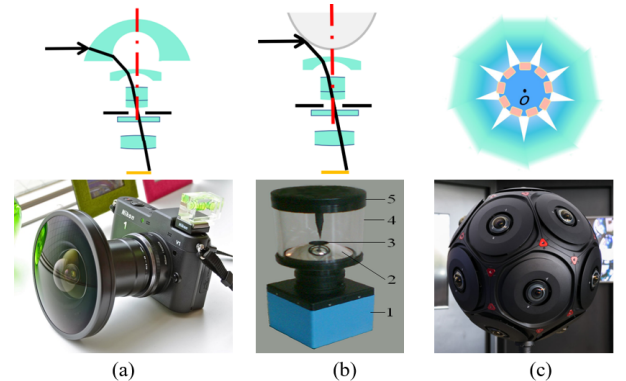


Fig. 3. The design principle of three commonly used spherical cameras. (a) dioptric camera; (b) catadioptric camera; and (c) polydioptric camera.

TABLE I

THE DETAILED CONFIGURATIONS OF WELL-KNOWN SPHERICAL CAMERAS. NOTICEABLY, THE RESOLUTION FOR LADYBUG5+ IS THE IMAGE DIMENSION OF AN INDIVIDUAL CAMERA.

Camera	Manufacturer	Principle	Number of lenses	Resolution(pixel)	Weight (kg)	Professional
Gear360	Samsung	polydioptric	2	4096×2048	0.13	×
Theta Z1	Ricoh	polydioptric	2	6720×3360	0.18	×
Theta X	Ricoh	polydioptric	2	11008×5504	0.17	×
Max 360	GoPro	polydioptric	3	5760×2880	0.16	×
Sphere	Insta360	polydioptric	2	6080×3040	0.19	×
Pro 2	Insta360	polydioptric	6	7680×7680	1.55	✓
Panono	Professional360	polydioptric	36	16000×8000	0.48	✓
Ladybug5+	Teledyne FLIR	polydioptric	6	2048×2464*	3.00	✓
Civetta	Weiss AG	dioptric	/	230 M	5.70	✓

sensors and image processing techniques, e.g., high-resolution digital cameras and high-precision stitching algorithms. Based on the design principle, spherical cameras can be divided into three major categories, i.e., dioptric cameras, catadioptric cameras, and polydioptric cameras [10], [19]:

- **Dioptric cameras** use a particular lens group to refract rays that compress the direction of the light entering the subsequent lens group. The obtained FOV reaches 360 degrees in the horizontal direction and is larger than 90 degrees in the vertical direction. Thus, two lenses combined back to back can capture the full surroundings. Fig. 3(a) is an example of the fisheye camera.
- **Catadioptric cameras** utilize the combination of a lens group for ray refraction and a special mirror for ray reflection, e.g., a parabolic, hyperbolic, or elliptical mirror, with a standard camera to achieve the FOV of 360 degrees and greater than 100 degrees in the horizontal and vertical direction, respectively. Compared with dioptric cameras, this design can reflect the surrounding light into the subsequent lens. Fig. 3(b) is an illustration of the catadioptric camera.
- **Polydioptric cameras** adopt multiple dioptric cameras to obtain a real spherical FOV, i.e., 360 degrees and 180 degrees in the horizontal and vertical directions, respectively, in which dioptric cameras have overlapping FOV to facilitate image stitching. Fig. 3(c) shows an example of a polydioptric camera comprising 16 cameras.

Among the three categories, polydioptric cameras have become the most extensively used spherical cameras for both professional and consumer-grade applications because of two main reasons. On the one hand, it can provide the full omnidirectional imaging technique; on the other hand, recorded images have extremely high resolutions due to the use of multiple cameras. Table I presents the detailed configurations of well-known spherical cameras, including the consumer-grade cameras, e.g., the Ricoh Theta series and Insta360 Sphere, and the professional cameras, e.g., Insta360 Pro2 and Teledyne FLIR Ladybug5+. Except for Weiss AG Civetta, all the other spherical cameras are designed by using the polydioptric mechanism. For consumer-grade cameras, the lens number is usually configured as 2 or 3. The illustration of the cameras is presented in Fig. 4. In general, there are two ways to increase the resolution of recorded images, i.e., increasing the number of integrated cameras or exchanging the

style of image recording. The former has been used in Panono and Ladybug5+, which enables instant image recording, e.g., equipped with an MMS system. The latter has been used in Weiss AG Civetta to obtain extremely high resolution. This strategy, however, sacrifices the capability of instant acquisition, and it is more suitable for site-based image recording.



Fig. 4. The illustration of the well-known spherical cameras. (a) Gear360; (b) Theta Z1; (c) Theta X; (d) Max 360; (e) Sphere; (f) Pro 2; (g) Panono; (h) Ladybug5+; (i) Civetta.

2) *Image representation*: Spherical images record the surrounding environments at each camera exposure position. In contrast to the 2D plane representation of perspective images, Fig. 5 presents the most widely used three types of image representation methods [20].

- The first one is the spherical representation. Objects in surrounding environments are mapped onto a sphere, as presented in Fig. 5 (a). Spherical representation is useful for panoramic navigation, which has been widely used